Cost-effectiveness of Ledipasvir/Sofosbuvir versus Pegylated Interferon/Ribavirin for chronic HCV in Egypt: a Markov model

Introduction

Chronic hepatitis C virus (HCV) is a blood-borne, progressive liver disease. HCV is often asymptomatic[1] until advanced stages, when patients may develop cirrhosis, liver failure and premature mortality. Beyond its effects on mortality and morbidity, chronic HCV is associated with substantial reductions in health-related quality of life and increasing economic burden as disease severity worsens. Egypt has experienced a particularly high burden of HCV with transmission being linked to unsafe, large scale parenteral anti-schistosomal therapy (PAT) campaigns between the 1950s and 60s. As a result of chronic HCV prevalence in the population, there is a need to identify and implement treatment strategies that are both clinically effective and economically viable.

Antiviral therapy aims to achieve a sustained virological response (SVR), eliminating the virus and reducing the risk of future HCV-related complications. Pegylated interferon plus ribavirin (PR) has been extensively used to treat genotype 4 HCV which is prevalent in Egypt. However, this treatment is associated with considerable side effects and relatively low SVR rates (~50%).[2] Direct acting antivirals (DAAs), including the combination of ledipasvir and sofosbuvir, achieves higher SVR rates and improves long-term health outcomes. However, DAAs are more expensive than PR which raises the question: do the additional health benefits gained by the use of DAAs justify the additional costs within a resource-strained health system?

In resource-limited settings, cost-effectiveness analysis is used to prioritise interventions that maximise health gains relative to available resources by comparing the incremental costs and possible health outcomes. As the clinical and economic consequences of HCV accumulate over decades, decision-analytic modelling is essential to extrapolate long-term outcomes beyond short-term clinical evidence. In this study a Markov cohort decision model was created in order to assess the cost effectiveness of ledipasvir/sofosbuvir in comparison to pegylated interferon/ribavirin for adults aged 40 years with moderate chronic HCV in Egypt. This analysis determines the lifetime costs, estimates quality-adjusted life years (QALYs) and explores uncertainties through sensitivity analysis. The findings will then be analysed to consider broader policy factors that can influence treatment coverage decisions.

Methods

A cost-effectiveness analysis (CEA) was conducted to compare the combination of pegylated interferon plus ribavirin (treatment A) with ledipasvir and sofosbuvir (treatment B) for the treatment of HCV in Egypt. A state-transition Markov cohort model was created in order to estimate the lifetime costs and quality adjusted life years (QALYs) gained from the perspective of the Egyptian Universal Health

Insurance Authority. This approach was adapted to reflect the progressive nature of hepatitis C infection which moves between clinically different disease states over time, meaning it is important to capture long term outcomes that extend beyond the duration of the trial. To account for this the model included the following health states: sustained virological response (SVR), moderate chronic hepatitis C, cirrhosis, liver failure and death. All patients entered the model in the moderate disease state at an average age of 40 years. The model used annual cycles, which was consistent with the availability of transition probability states. The model also assumes a memoryless process where transition probabilities depend solely on current health state and does not factor previous disease history. The Markov model structure is shown in Figure 1.

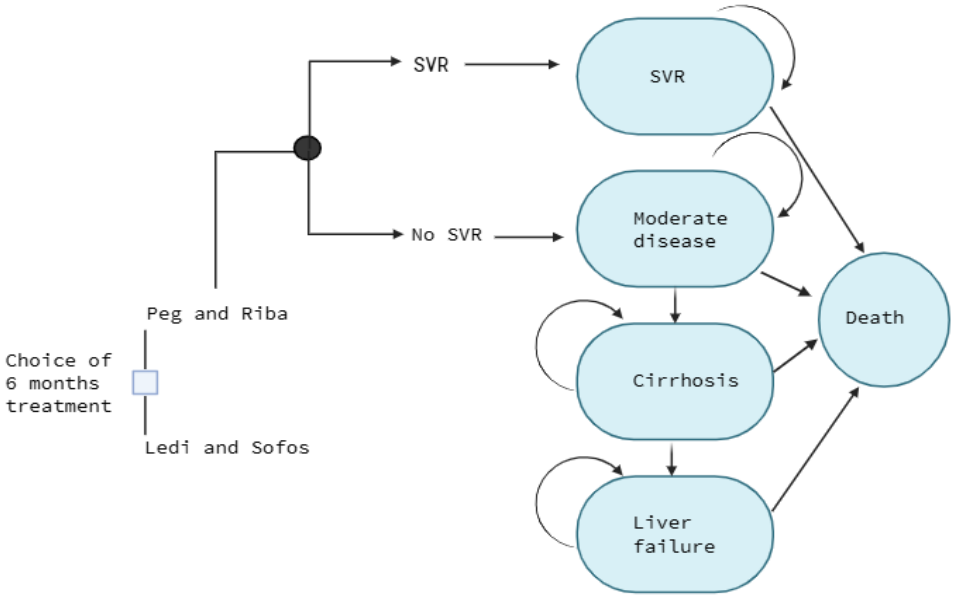


Figure 1: Markov model structure representing disease progression pathways for chronic hepatitis C [3]

Both treatment regimens (A and B) were assumed to be administered for six months. Treatment effectiveness was determined by the probability of acquiring a sustained virological response (SVR) following 6 months of treatment. The probability of SVR was taken as being 39.5% (0.395) for treatment A and 89.0% (0.89) for treatment B. SVR was modelled as an absorbing health state, representing sustained viral clearance and minimal risk of further disease progression. For patients who did not achieve SVR, disease progression followed annual transition probabilities from moderate disease to cirrhosis and from cirrhosis to liver failure. Disease progression is assumed to be irreversible, which reflects clinical evidence that advanced liver damage is non-reversible in many cases[4]. However, this assumption may overestimate the potential long term disease burden. All patients faced age specific all-cause mortality risks based on general population life tables. Additional excess mortality risk was applied to patients in the liver failure health state.

The analysis adopted the perspective of the Universal Health Insurance Authority (UHIA)[5] and included the direct medical costs only. The UHIA serves as the single purchaser of curative health services for the entire population and aims to allow equitable access to quality health services whilst ensuring financial protection[5]. Costs were expressed in Egyptian pounds (E£) and were adjusted to a common price year (2023). The treatment costs reflected the negotiated discounted price of treatment B and the cost of treatment A. In addition, annual health-state costs associated with moderate disease, cirrhosis and liver failure were included. Patients that achieved an SVR were assumed to no longer accumulate further disease related costs. In this study, indirect costs were excluded due to data limitations.

The health outcomes of the study are measured in quality-adjusted life years (QALYs). Health-state utility values were derived from EQ-5D-3L data that were reported for patients at different stages of HCV infection. In the absence of Egyptian-specific utility estimates, values from a UK population were used. QALYs were calculated by weighting time spent in each health state by the corresponding utility value. A lifetime time horizon was implemented in order to capture the long-term costs and disease progression. Both costs and health outcomes were discounted at an annual rate of 3.5% in line with standard economic evaluation practice.

Deterministic one-way sensitivity analyses were conducted to assess the impact of uncertainty in key model parameters, including SVR probabilities, disease progression rates, health-state costs, utility values and discount rates. Results were summarised using a tornado diagram to identify parameters with the greatest influence on the incremental cost- effectiveness ratio (ICER).

An additional probabilistic sensitivity analysis was undertaken to assess joint parameter uncertainty. Key model parameters were assigned probability distributions based on their underlying characteristics, with beta distributions used for probabilities and utilities and gamma distributions for costs. A Monte Carlo simulation with 1,000 iterations was performed to generate distributions of costs, QALYs and ICERs.

Results and Analysis

The base-case analysis demonstrates that treatment with Ledipasvir and Sofosbuvir (Treatment B) is associated with higher total costs. However, it is also associated with substantially better health outcomes when compared with pegylated interferon plus ribavirin (Treatment A). Specifically, treatment B generated an additional 1,278 QALYs and 697 life years at an incremental cost of approximately £14.96 million for the cohort discussed. The resulting incremental cost-effectiveness ratio (ICER) of £11,704 per QALY gained shows that the additional health benefits are achieved at a relatively low additional cost per unit of health outcome.

Under base-case assumptions, ledipasvir and sofosbuvir can be considered cost effective as the ICER falls below the willingness to pay (WTP) threshold of £34,580 per QALY. From an economic perspective, the additional health gains outweigh the resource requirements.

Outcome	Pegylated interferon + ribavirin (A)	Ledipasvir + Sofosbuvir (B)	Incremental (B – A)
Total Costs (E£)	£ 34,072,928	£ 49,031,442	£ 14,958,513
Total QALYs	11176.71	12454.77	1278.05
Total Life Years	17089.11	17786.88	697.77
ICER (E£/QALY)	—	—	**11,704**

Table 1: Base case analysis

Deterministic sensitivity analysis was undertaken to identify which input parameters impacted the model outcome the most. A tornado diagram was constructed which showed that the model outcomes were most sensitive to health state utility parameters, especially the utility assigned to sustained virological response (SVR), and to the discount rate. This indicates that the economic value of ledipasvir and sofosbuvir is primarily driven by long-term health gains brought around by successful treatment as opposed to cost inputs. The strong influence of discounting reflects the long-term nature of hepatitis C infection and its progression, as interventions that have higher upfront costs generate benefits over long periods of time. Parameters related to disease progression such as transition probabilities to cirrhosis (tpCirrhosis)and the probability of achieving sustained virological response for both treatment strategies (pSVR_LS and pSVR_PR) have an influence on the results. However, these have a smaller influence on the results than the utility assigned to SVR. The deterministic sensitivity analysis is shown in Figure 2.

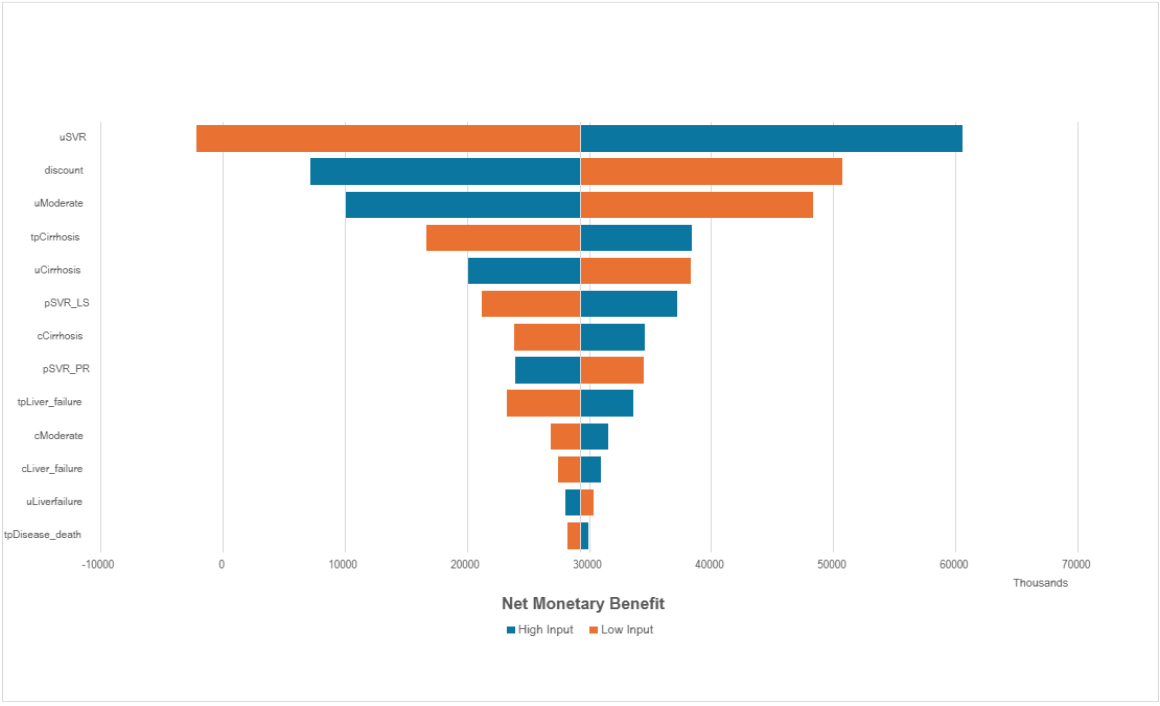


Figure 2: Tornado Diagram

Probabilistic sensitivity analysis was undertaken to further test the robustness of the findings by varying each parameter simultaneously and further supported the cost-effectiveness of ledipasvir and sofosbuvir. The cost effectiveness plane shown in Figure 3 displays the distribution of incremental costs and QALYs across all the simulations which shows that results remain consistently associated with improved health outcomes even with the uncertainty.

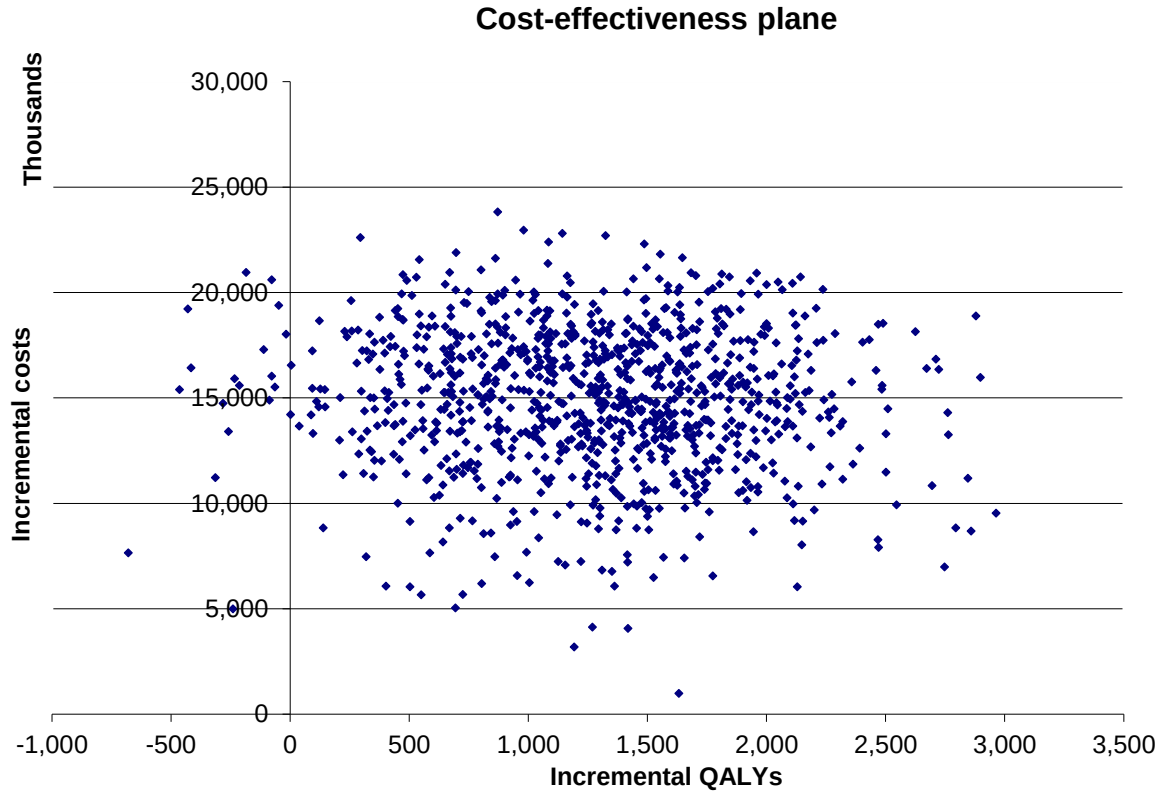


Figure 3: Cost-effectiveness plane

The cost effectiveness analysis curve in Figure 4 further quantifies decision uncertainty as it shows the probability of ledipasvir and sofosbuvir being cost effective across different willingness-to-pay thresholds. The curve increases substantially as the willingness to pay threshold (£34,580) rises. The willingness-to-pay threshold was derived using a GDP-based approach, representing approximately 0.39 times GDP per capita which for Egypt is £88,665 Egyptian pounds, resulting in a threshold of £34,580 per QALY. At the threshold, the probability of cost-effectiveness exceeded 90%, which indicates high decision confidence.

Outcome	Mean Incremental Cost	Mean Incremental QALY	Probability CE at £34,580
PSA results	34072928	1269	0.91

Table 2: Probabilistic sensitivity analysis table

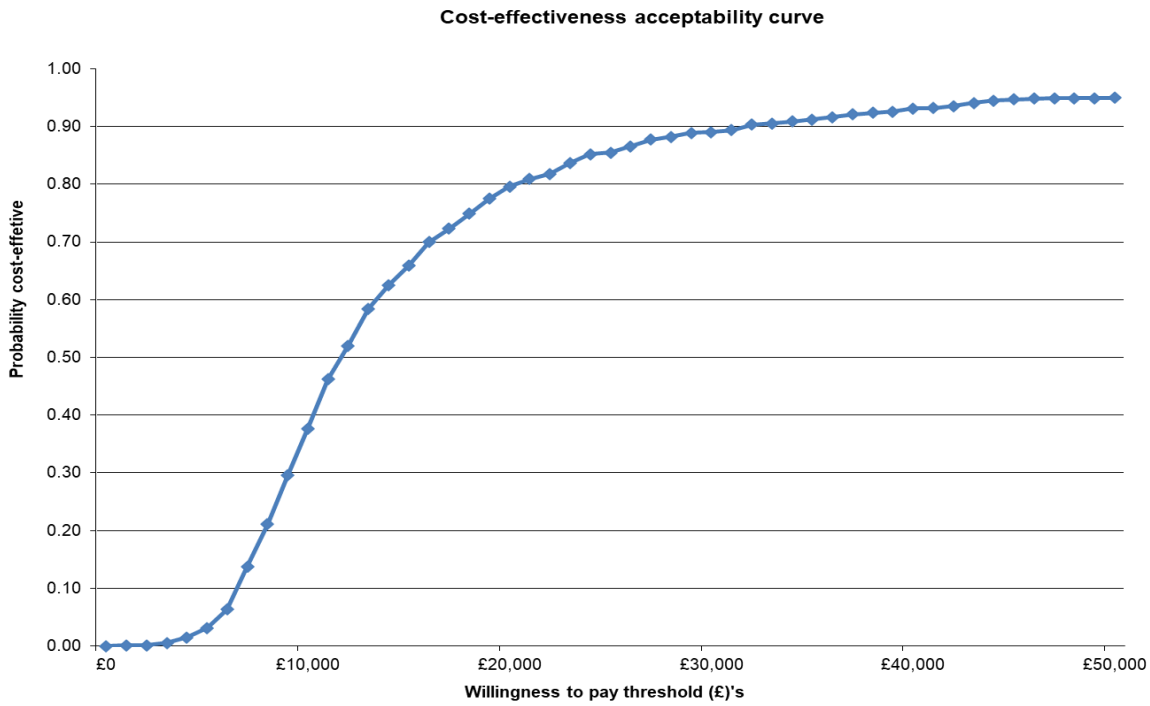


Figure 4: Cost-effectiveness acceptability curve

Overall, base-case and sensitivity analyses indicate that ledipasvir and sofosbuvir represents good value for money, with a high probability of cost-effectiveness at the willingness to pay threshold (£34,580).

Discussion

The base case analysis suggests that Ledipasvir and Sofosbuvir is cost-effective when compared with pegylated interferon and ribavirin for patients with chronic hepatitis C at the moderate disease stage, as it generates substantial incremental health gains at an ICER of £11,704 per QALY gained. This falls well below the WTP threshold of £34,580. The consistency of findings across deterministic and probabilistic sensitivity analyses strengthens the confidence vested in the outcome. However, interpretation of these results must take into account the underlying model assumptions and broader policy considerations. The primary driver of cost-effectiveness for treatment B is the increased sustained virological response (SVR) rate. By achieving SVR it reduces the likelihood of progression to cirrhosis and liver failure, which generates significant gains in survival and quality adjusted life expectancy. Therefore, although upfront treatment costs are higher for Ledipasvir and Sofosbuvir, these are offset by significant decreases in downstream morbidity and health related quality of life. Despite the intervention having a higher acquisition cost, it still represents an efficient use of healthcare resources when the long-term benefits are taken into account.

Deterministic sensitivity analysis demonstrated that the outcomes that were the most sensitive to utility values were associated with SVR and moderate disease, as well as the discount rate. This was expected due to the lengthy nature of hepatitis C progression and this displays that the model is more reliant on long-term quality of life gains as opposed to immediate cost savings. It was found that varying key results in important parameters did not result in ICER values going beyond the willingness to pay threshold. This indicates that the base-case conclusion is quite robust.

Probabilistic sensitivity analysis (PSA) further supported the robustness of the findings. The cost-effectiveness plane indicated that the majority of simulations fell within the northeast quadrant, which reflects increased costs alongside improved health outcomes. The cluster of simulations to the right of the origin suggests that the incremental QALYs stay positive across a wide range of parameter uncertainty. Additionally, the cost-effectiveness acceptability curve displayed a high probability that Ledipasvir and Sofosbuvir is cost-effective at the specified willingness-to-pay threshold. This indicates relatively low decision uncertainty and enables the strengthening of confidence in the overall conclusion by accounting for the uncertainty that existed in multiple parameters rather than relying on point estimates alone.

In spite of these strengths, several methodological limitations can be considered. Firstly, the model assumes that achieving SVR represents a permanent health state without risk of reinfection or possible disease progression. This can undermine the possible long-term benefits. Secondly, utility values were taken from UK populations and may not fully encapsulate health preferences of the Egyptian population, this can impact QALY estimates and damage the credibility of the findings. Finally, the lack of direct randomised controlled trial evidence causes uncertainty regarding comparative effectiveness between treatment strategies and may not capture the full heterogeneity of disease progression and patient treatment outcomes.

From a policy standpoint, the findings indicate that the uptake of Ledipasvir and Sofosbuvir can generate population health gains at an acceptable cost per QALY. However, cost-effectiveness doesn't necessarily equate to affordability. The higher acquisition cost of the intervention could present budgeting challenges for healthcare systems with restricted resources, especially those of which have a high prevalence of hepatitis C such as Egypt. Decision makers such as the UHIA in this case should consider both financial feasibility and economic efficiency when assessing implementation strategies.

Conclusion

To conclude, the combined evidence from the base-case analysis and deterministic and probabilistic analyses indicates that the combination of Ledipasvir and Sofosbuvir is likely to represent a cost-effective intervention under a large range of plausible assumptions. In order to strengthen the evidence,

future research should incorporate locally derived utility values and utilise real-world effectiveness data if it is available.

References

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